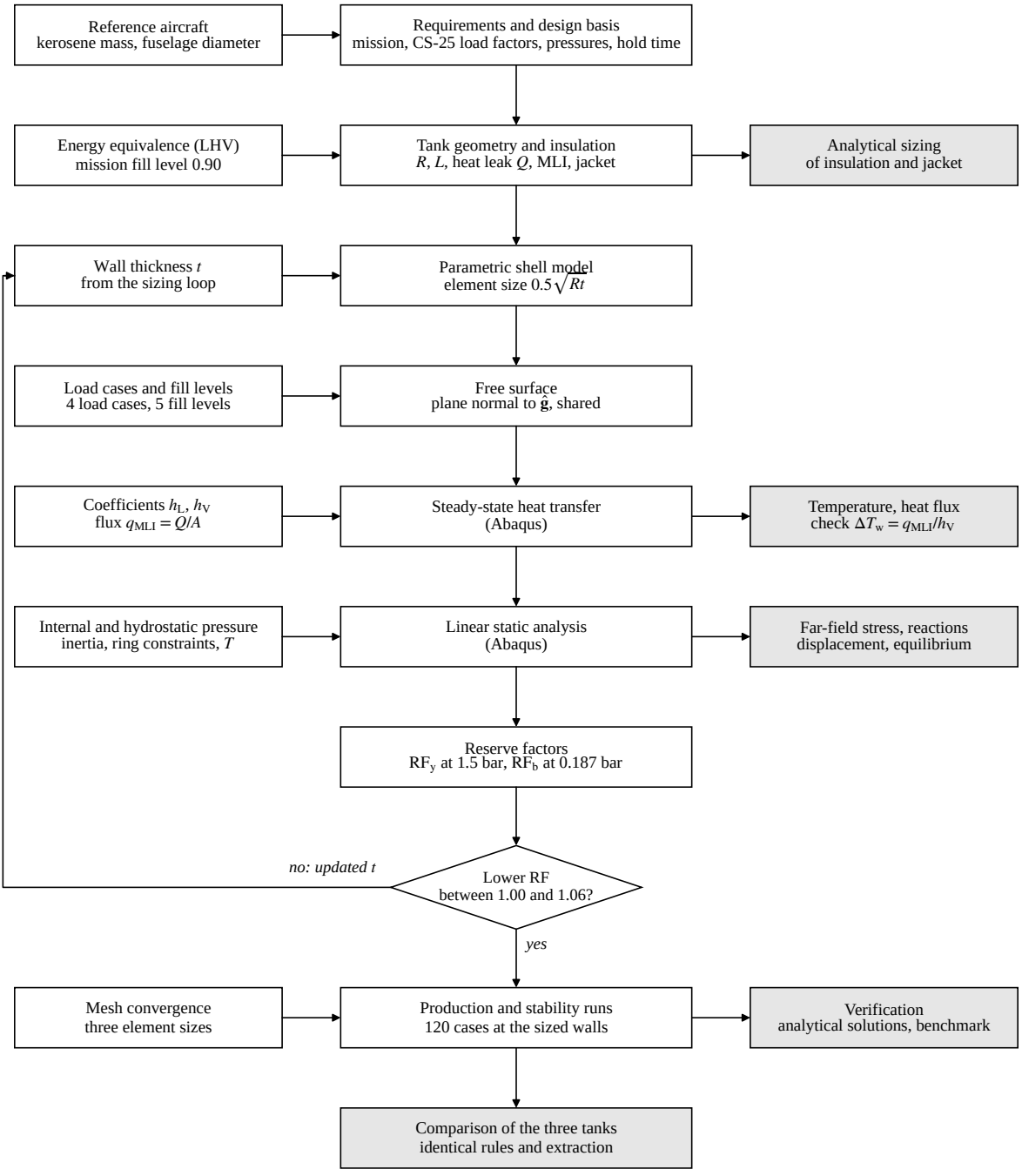
Reference aircraft
kerosene mass, fuselage diameter
Requirements and design basis
mission, CS-25 load factors, pressures, hold time
Energy equivalence (LHV)
mission fill level 0.90
Tank geometry and insulation
R, L, heat leak Q, MLI, jacket
Analytical sizing
of insulation and jacket
Wall thickness t
from the sizing loop
Parametric shell model
element size 0.5√Rt
Load cases and fill levels
4 load cases, 5 fill levels
Free surface
plane normal to ĝ, shared
Coefficients h_L, h_V
flux q_MLI = Q/A
Steady-state heat transfer
(Abaqus)
Temperature, heat flux
check ΔT_w = q_MLI/h_V
Internal and hydrostatic pressure
inertia, ring constraints, T
Linear static analysis
(Abaqus)
Far-field stress, reactions
displacement, equilibrium
Reserve factors
RF_y at 1.5 bar, RF_b at 0.187 bar
no: updated t
Lower RF
between 1.00 and 1.06?
yes
Mesh convergence
three element sizes
Production and stability runs
120 cases at the sized walls
Verification
analytical solutions, benchmark
Comparison of the three tanks
identical rules and extraction